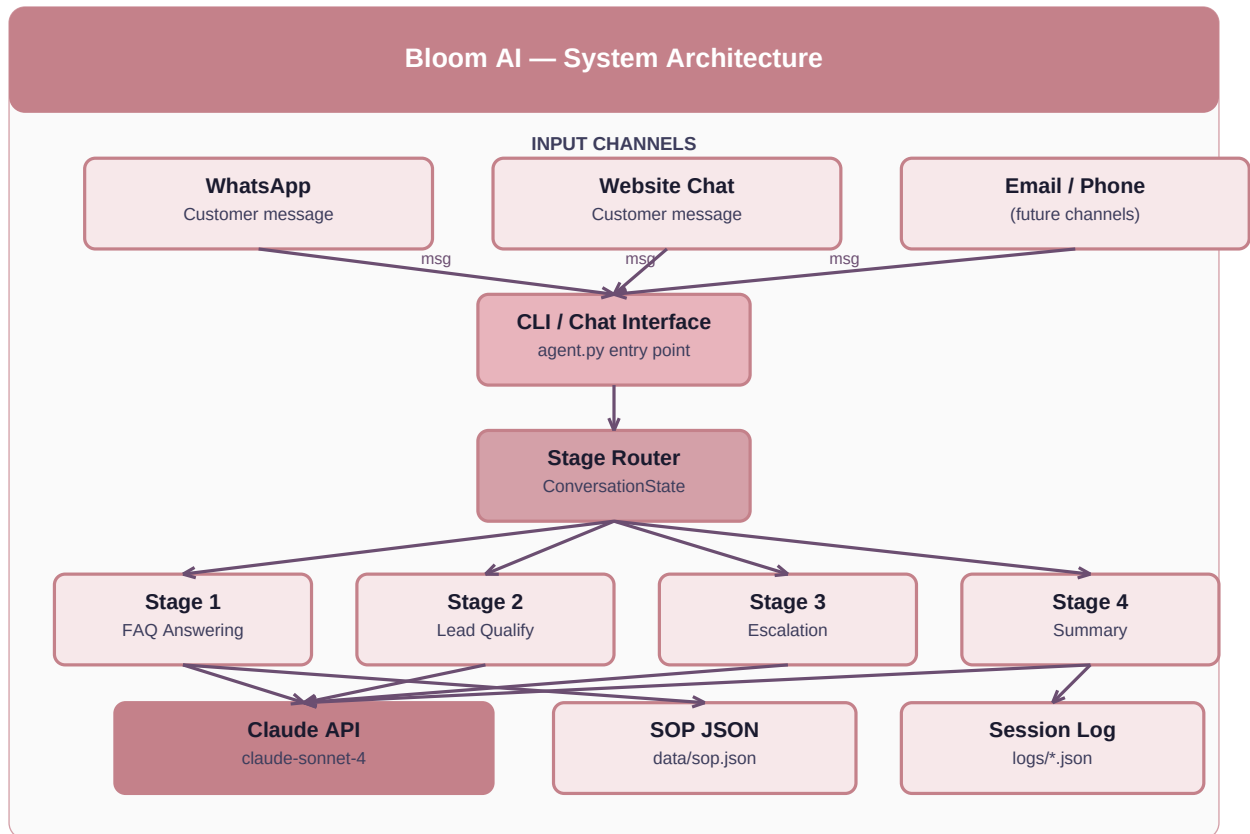


Bloom AI

Customer Support Agent — System Architecture

Closira AI Engineering Intern Assignment

1 · High-Level Architecture Diagram



2 · Workflow Stages

#	Stage	Entry Condition	Key Logic	Exit Condition
1	FAQ Answering	Any customer message in default state	System prompt + SOP injected. Model answers only from SOP. Unanswered Q counter tracked.	Answer delivered or unanswered count ≥ 2
2	Lead Qualification	/qualify command or auto-trigger after 3 FAQ turns	QUALIFICATION_SYSTEM_PROMPT. One question at a time. [QUALIFICATION_SUMMARY] parsed.	All 3 dimensions collected
3	Escalation Detection	[ESCALATE:] tag in response OR unanswered count ≥ 2 OR sentiment/complaint	stage = 'escalated'. AI responses stop. Reason logged to session JSON.	Human agent takes over

4	Conversation Summary/summary or /quit command	SUMMARY_SYSTEM_PROMPT. Full history passed as context. 5-field structured output.	Summary displayed + log written to disk
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3 · Escalation Detection (Two-Layer)

Layer	Mechanism	Trigger Conditions
Model-side	[ESCALATE: reason] tag embedded in response	Complaint · Medical question · Pricing negotiation Angry/distressed tone · Explicit human request Low confidence (cannot confirm from SOP)
System-side	unanswered_count ≥ 2 (ConversationState counter)	2 consecutive responses containing uncertainty phrasing — fires even if model omits the tag

4 · Hallucination Prevention Strategy

Layer	Implementation
SOP Injection	Full data/sop.json embedded into system prompt at runtime. Model has direct access to every fact it is permitted to state — no reliance on parametric memory.
Hard Rule	"ANSWER ONLY FROM THE SOP ABOVE. Never invent services, prices, staff names, availability, or policies." — Categorical constraint covers the most common hallucination vectors.
Confidence Fallback	If not fully confident, the model is instructed to say so and escalate rather than guess. Appends [ESCALATE: Low confidence — question outside SOP scope].

5 · Data & State Flow

Object	Type	Purpose
ConversationState	Python dataclass	Holds messages, stage, unanswered_count, escalation_reason, qualification_data, session_id
state.messages	list[dict]	Full OpenAI-format message history — passed to every API call for context continuity
state.turn_log	list[dict]	Flat audit log with stage annotation per turn — written to logs/session_<id>.json on exit
qualification_data	dict	Parsed from [QUALIFICATION_SUMMARY] tag; merged into final summary prompt
escalation_reason	str None	Extracted from [ESCALATE: ...] tag; recorded in session log and summary

6 · Project File Map

File / Path	Purpose
agent.py	Main entry point. CLI loop, stage router, all prompt calls.
data/sop.json	Knowledge base. The only source of truth for FAQ answers.

prompt_design.md	Full prompt documentation: design decisions, hallucination prevention, escalation logic, tone/persona rationale.
test_transcripts/01-05	One markdown file per required test scenario.
logs/session_*.json	Auto-generated per session. Stores full turn log, stage transitions, escalation reason, qualification data.
README.md	Setup instructions, dependency list, trade-offs.

7 · Technology Stack

Component	Technology	Notes
AI Model	claude-sonnet-4-20250514	Via Anthropic Python SDK
Prompting Strategy	Multi-system-prompt	Separate prompts per stage; SOP injected at runtime
Escalation Detection	Regex + counter	extract_escalation() parses [ESCALATE:] tag
State Management	In-memory Python class	ConversationState; stateless across sessions
Session Logging	JSON file	logs/session_<timestamp>.json
Interface	Python CLI	No frontend required per brief
Dependencies	anthropic	Standard library only (json, re, os, datetime)